

Airframe Technology Master

7 - DESIGN

Lecture 2

Configuration Management

Rubén Cubero

May 14th, 2025

Who am I?

Rubén Cubero-Muñoz, Head of Design Config Sp , TADIM1

Personal Data

1982, father of two tiny women and husband of a beautiful wife

Resident in Madrid, Madrilenian by birth and heart

Atlético de Madrid fan



Education

Master Engineer in Aeronautics in Universidad Politecnica de Madrid (2006)

Erasmus @ Politecnico di Milano

Work Experience

From 2011 in Design Configuration Management department, since 2017 Team leader of the group.

Before joining Airbus, experience in civil Engineering in Dragados (Building Company)

Main projects involved:

- Electrical process final integration in LTA and MRTT
- A400M DASS UK operational support
- A400M PASS SSI (PASS V3) deployment
- Export Control requirements inclusion in PLM

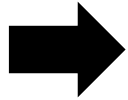
Rationale for Configuring a product

Space X launch site of Falcon 9



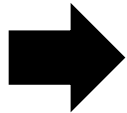
5 Ws on configuration Management

WHAT?



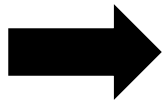
CM is a process oriented to **establish and maintain the consistency** of the functional, physical and performance **characteristics of the product**, compared with its operational requirements and its final design through **the whole life cycle**.

WHO?

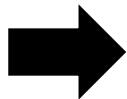


Actors from all are industrial companies and different functions intervene in CM process. The Configuration Management departments are those keeping an eye on the proper function of the process but do not populate the Configuration of the product.

WHEN?

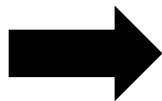


CM is performed **through all the life cycle** of the product. Depending on the product, different phases will be highlighted. In Design, Development and Production, the main phase is the development one up to product delivery.

WHERE/
HOW?

CM is a process that is included in **ad-hoc tools** register and consult the configuration for those tools serving as a base for many other tools. CM is a process and its tooling is generated to reproduce the process.

WHY?



CM is a process performed to register in a designated way the product (not depending in user goodwill) with the intention of being able to retrieve it and know why and when the changes in it were made. And also, because **it is a strong requirement from the authorities**.

CREATE

MODIFY

RETRIEVE

Configuration Management in EASA part 21

21.A.31 Type design

Regulation (EU) No 6/2002

(a) The type design shall consist of:

1. the drawings and specifications, and a listing of those drawings and specification necessary to define the configuration and the design features of the product shown in conformity with the applicable type certification basis and environmental protection requirements;
2. information on materials and processes and on methods of manufacture and assembly of the product necessary to ensure the conformity of the product;
3. an approved airworthiness limitations section and instructions for continued airworthiness as defined by applicable certification specifications; and
4. any other data necessary to allow a comparison and determination of the characteristics of noise, fuel venting, and exhaust emissions (where applicable) of

8. Aircraft configuration as relevant for the permit to fly

- 8.1 The above aircraft for which a permit to fly is requested is defined in [add reference to the document(s) identifying the configuration of the aircraft. Same as required in [AMC 21.A.263\(d\)\(6\)](#) or [AMC 21.A.709\(b\)](#) application approval form 18A or 18B, box 6]
- 8.2 The aircraft is in the following situation [state the maintenance schedule:

[Describe status]

BASELINING

21.A.33 Inspection and tests

Regulation (EU) No 748/2012

(a) The applicant shall perform all inspection and tests necessary to demonstrate compliance with the applicable type certification basis and environmental protection requirements.

(b) Before each test required by point (a) is undertaken, the applicant shall have determined:

1. for the test specimen:
 - (i) that materials and processes adequately conform to the specifications for the proposed type design;
 - (ii) that parts of the products adequately conform to the drawings in the proposed type design;
 - (iii) that the manufacturing processes, construction and assembly adequately conform to those specified in the proposed type design; and
2. that the test equipment and facilities used for the test are appropriate and calibrated.

7. Control of aircraft configuration

ED Decision 2012/020/R

It shall be with a method of the control of any change or repair made to the aircraft, and repairs that do not meet the conditions established to the permit to fly. Changes should be approved in accordance with [21.A.713](#) and when necessary a new permit be issued in accordance with [21.A.711](#).

CHANGE MANAGEMENT

AMC No 2 to 21.A.31 (c) Obligations of the manufacturer
Conformity with Applicable Design

Individual configurations are often based on the needs of the customer and improvements which may be introduced by the type certificate holder. There are also likely to be unintended divergences (conformances or non-conformances) during the manufacturing process. All these are required to have been approved by the design approval applicant/holder, or when not the Agency.

SUPPLIER CONTROL

ED Deci

21.A.34 Retention of information

Regulation (EU) No 6/2002

All relevant design information, drawings and test reports, including inspection records for the product tested, shall be held by the manufacturer or its authorised representative until the disposal of the Agency and shall be retained in order to provide the information necessary to ensure the continued airworthiness, continued validity of the operational suitability data and compliance with applicable environmental protection requirements of the product.

INFORMATION RETRIEVAL

21.A.31 Type design

(a) The type design shall consist of:

1. The drawings and specifications, and a listing of those drawings and specifications, necessary to define the configuration and the design features of the product shown to comply with the applicable type certification basis and environmental protection requirements (where applicable);
2. Information on materials and processes and on methods of manufacture and assembly of the product necessary to ensure the conformity of the product;
3. An approved airworthiness limitations section of the instructions for continuing airworthiness as defined by the applicable airworthiness codes; and
4. Any other data necessary to allow by comparison, the determination of the airworthiness, the characteristics of noise, fuel venting, and exhaust emissions (where applicable) of later products of the same type.

(b) Each type design shall be adequately identified.

21.A.33 Investigation and tests

(a) The applicant shall perform all inspections and tests necessary to show compliance with the applicable type certification basis and environmental protection (where applicable).

(b) Before each test required by paragraph (a) is undertaken, the applicant shall have determined:

1. For the test specimen:
 - i. That materials and processes adequately conform to the specifications for the proposed type design;
 - ii. That parts of the products adequately conform to the drawings in the proposed type design;
 - iii. That the manufacturing processes, construction and assembly adequately conform to those specified in the proposed type design; and

that the test equipment and all measuring equipment used for tests are adequate for the test and are appropriately calibrated.

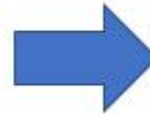
21.A.55 Record keeping

All relevant design information, drawings and test reports, including inspection records for the product tested, shall be held by the type certificate or restricted type certificate holder at the disposal of the Authority and shall be retained in order to provide the information necessary to ensure the continued airworthiness and compliance with applicable environmental protection requirements (where applicable) of the product.

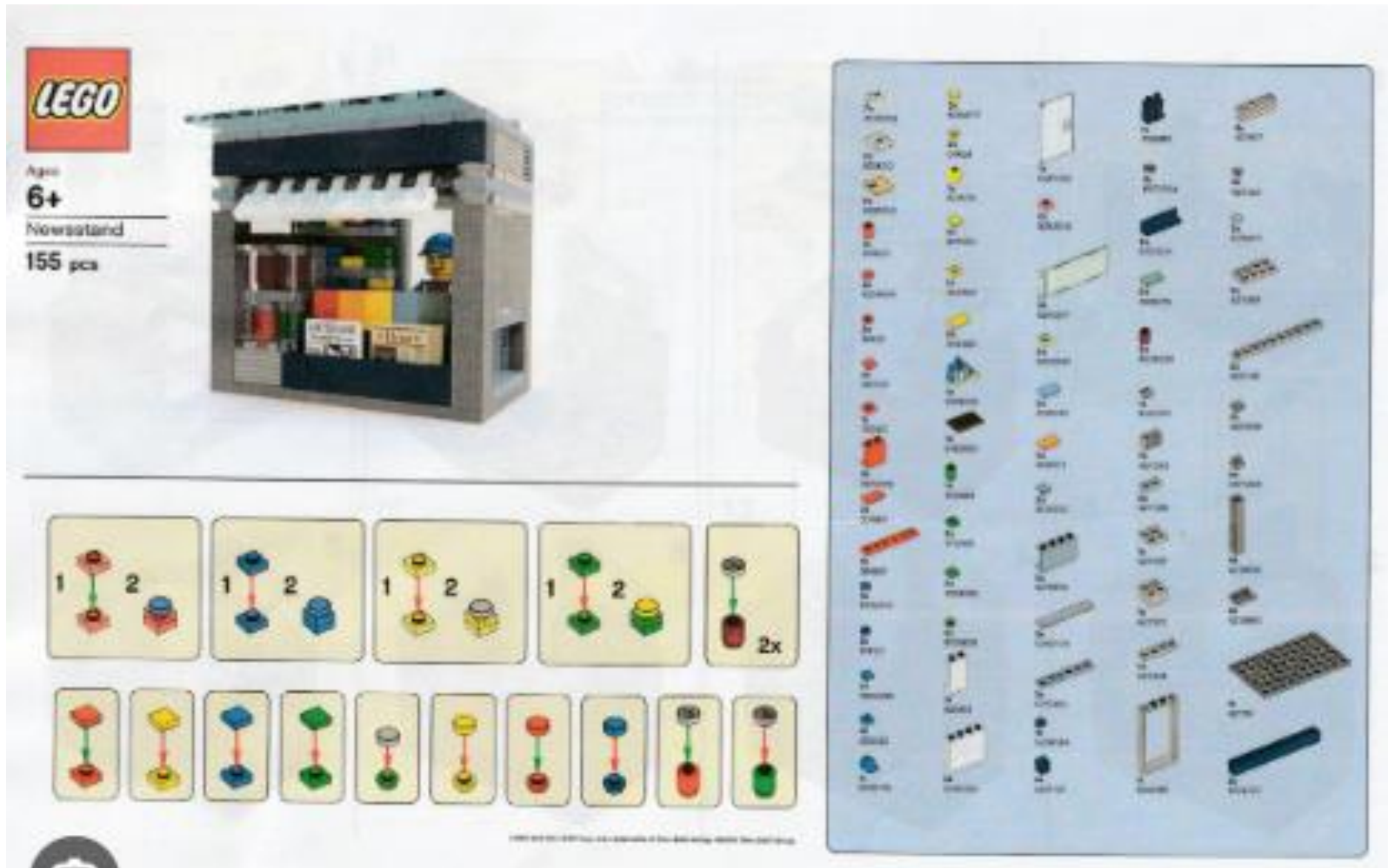
The Lego example: Create



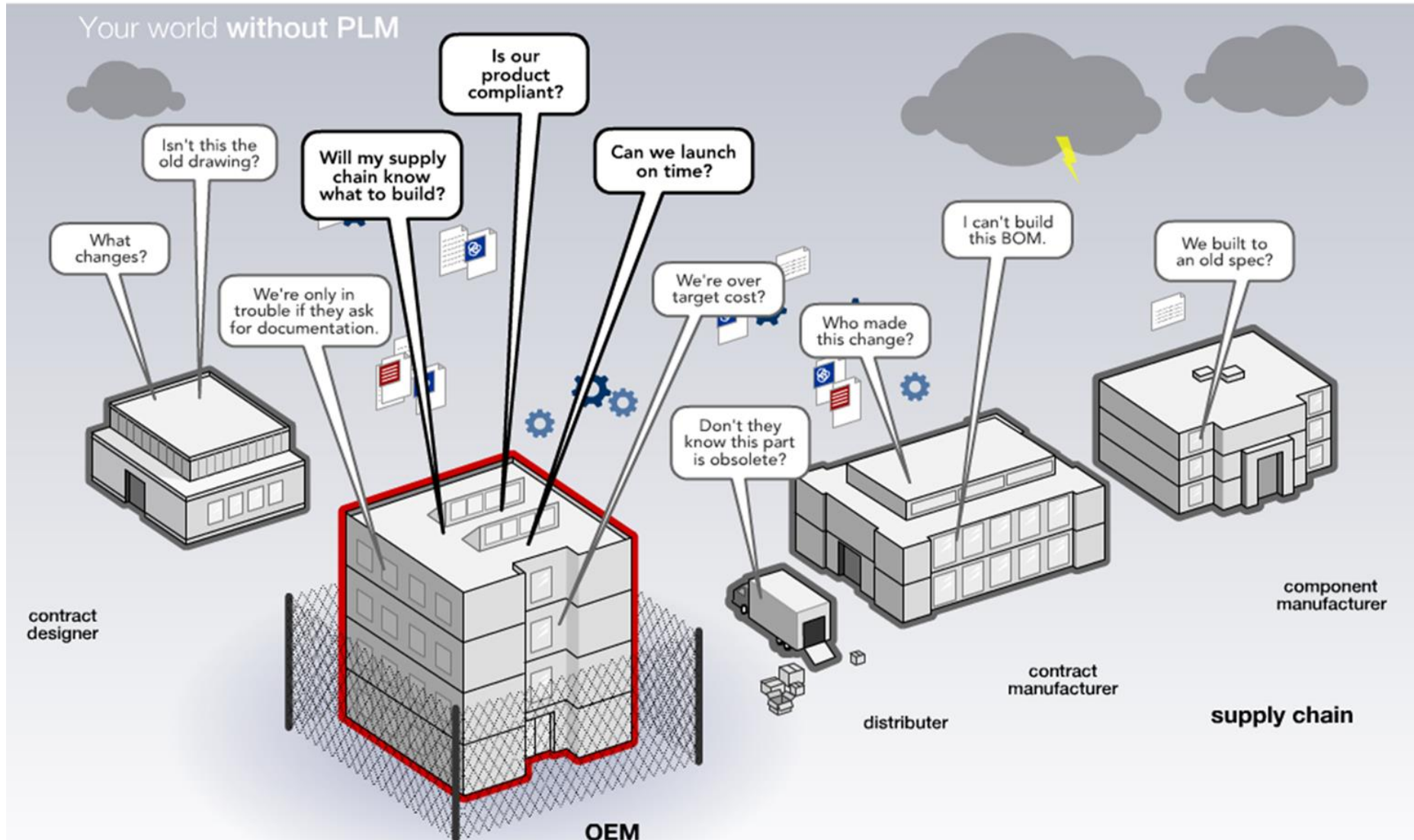
The Lego example: Modify



The Lego example: Retrieve



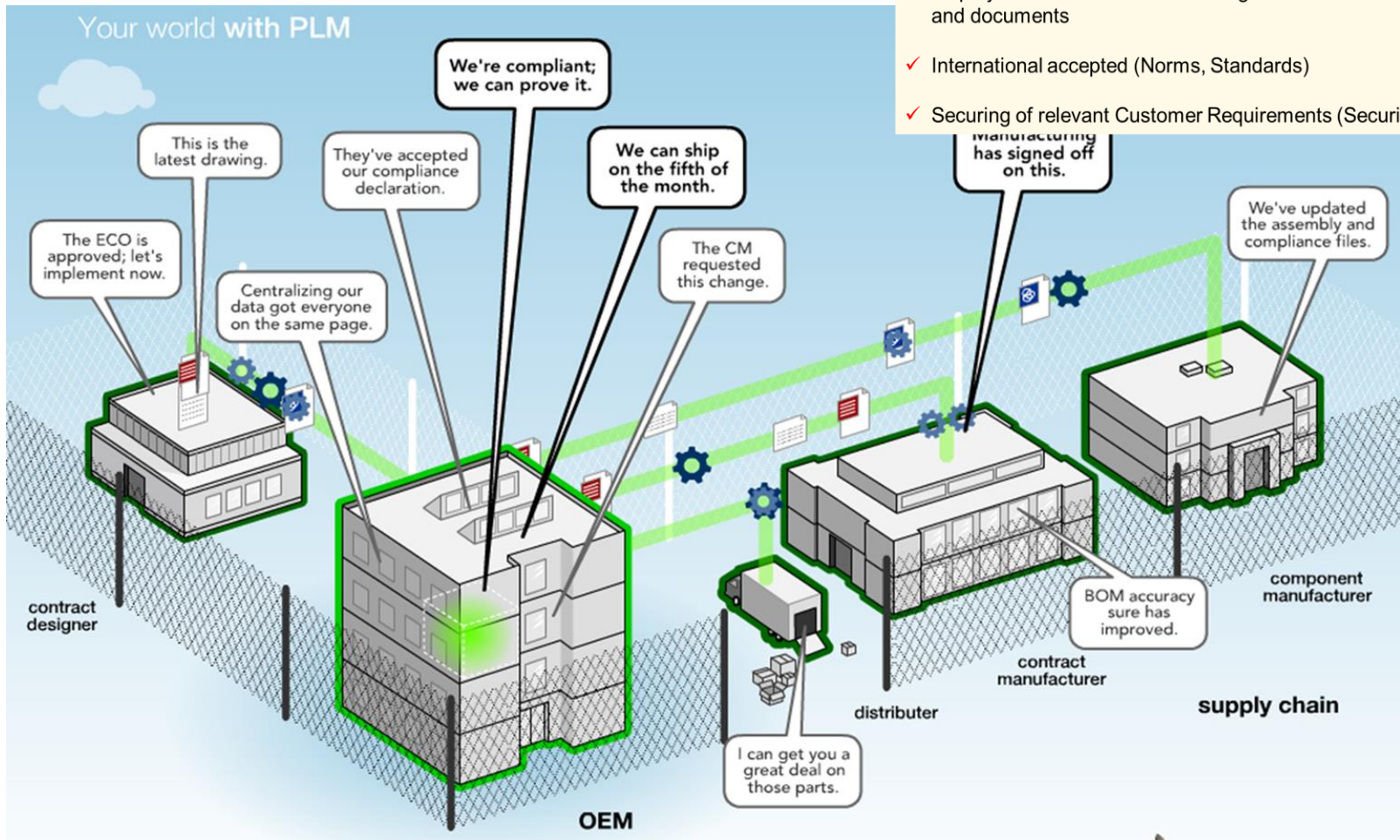
Configuration Management purpose



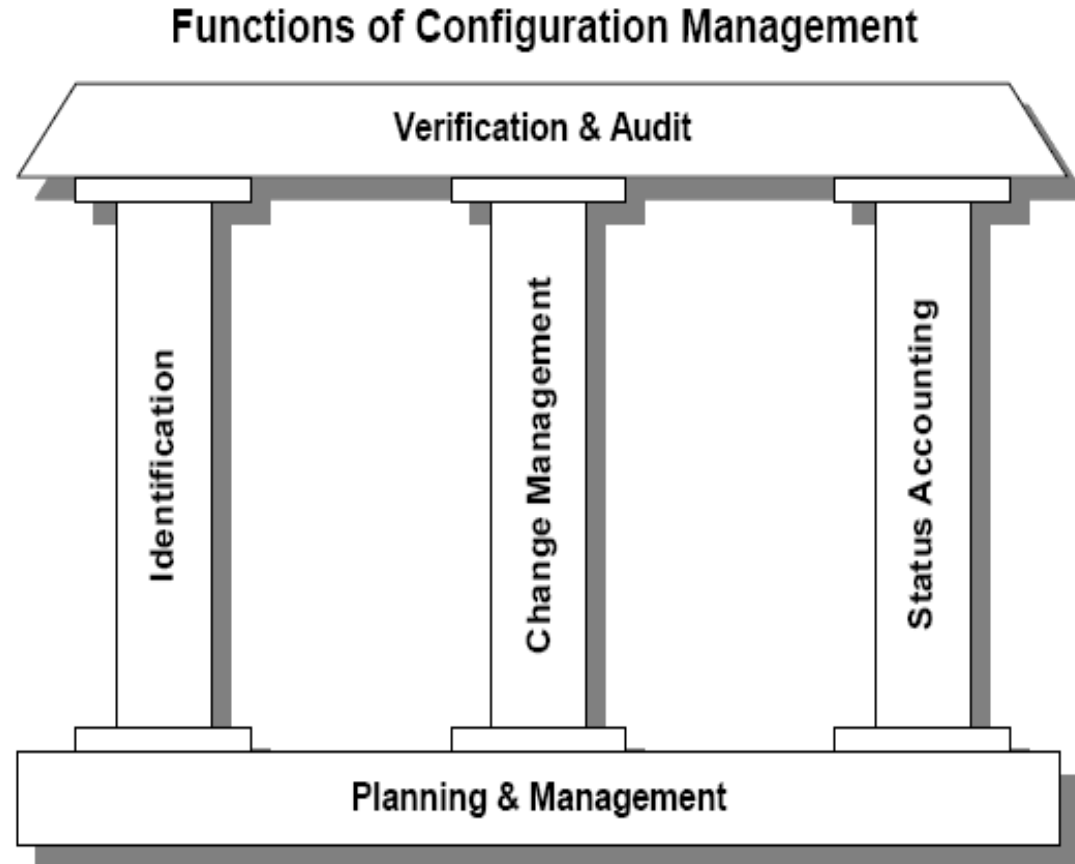
Configuration Management

purpose

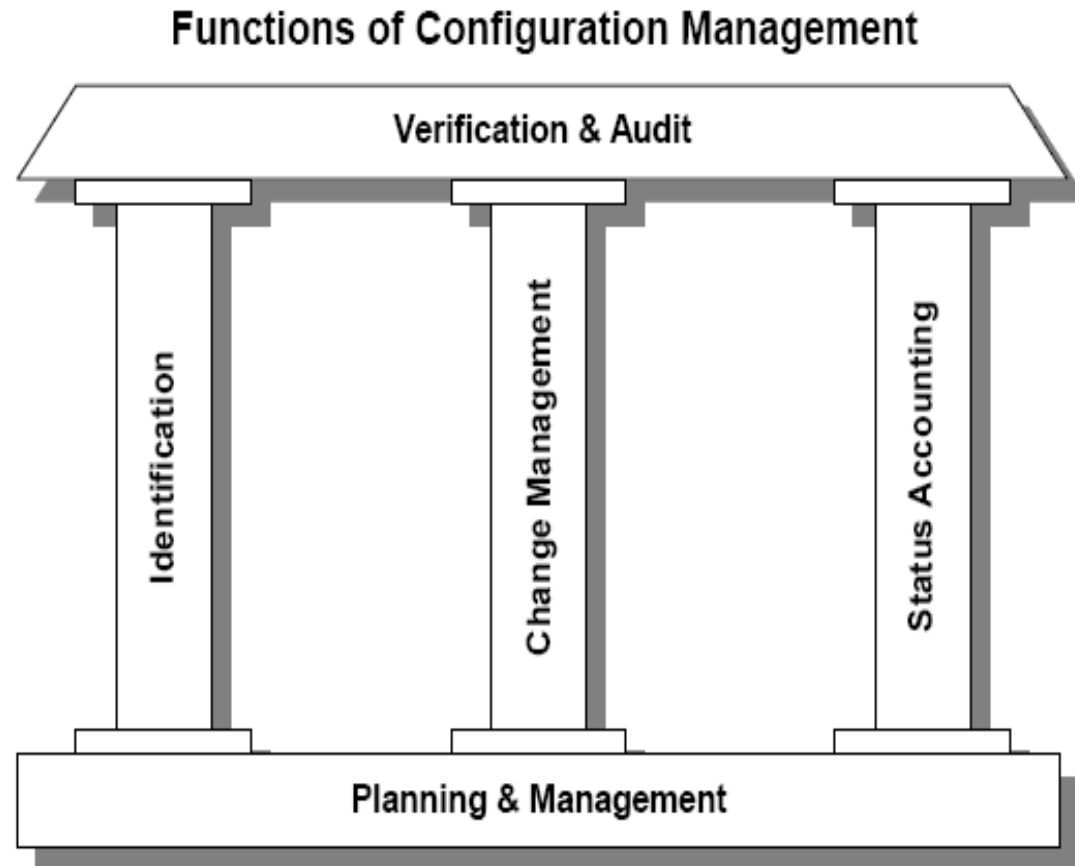
- ✓ Coordination of processes (Know how of processes and interfaces)
- ✓ Traceability of project state (Reporting)
- ✓ All project members work according to the same released processes and documents
- ✓ International accepted (Norms, Standards)
- ✓ Securing of relevant Customer Requirements (Security rules,)



1. CM Planning and Management
2. Configuration Identification
3. Configuration Change Management
4. Configuration Status Accounting
5. Configuration Verification and Audit



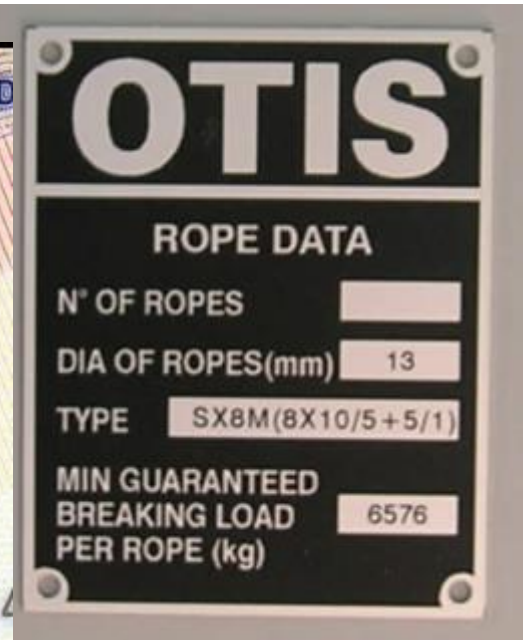
1. CM Planning and Management
- 2. Configuration Identification**
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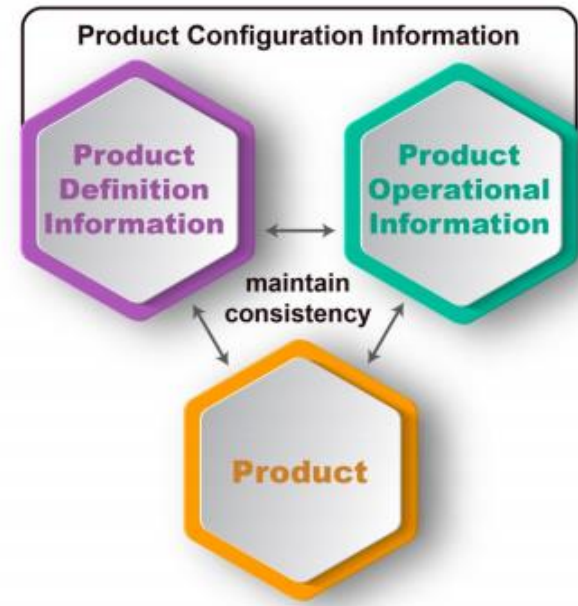
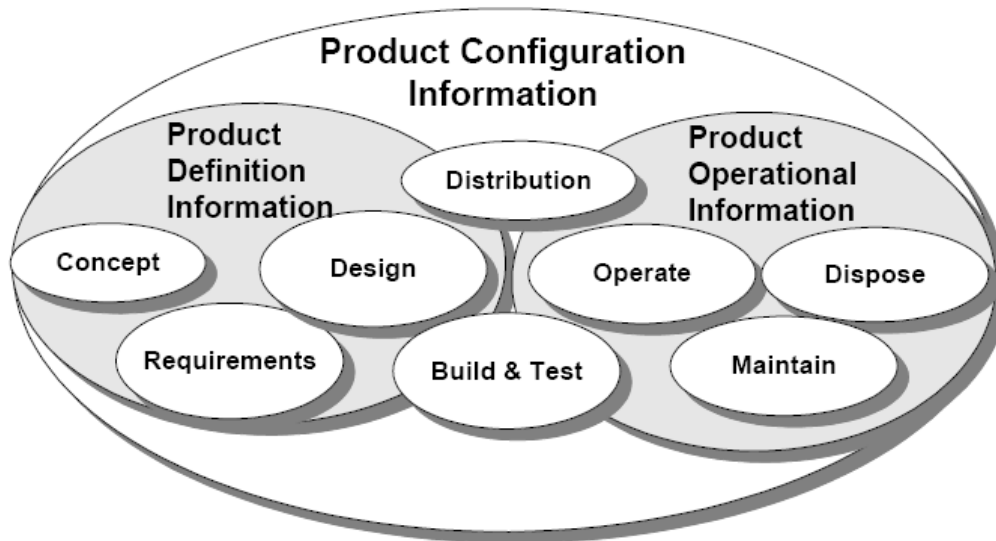
HIBOUTIK

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Descubra su regalo escaneando este QR código





The Product Definition Information describes the product performances and its physical and functional attributes, from the spec to its definition documentation.

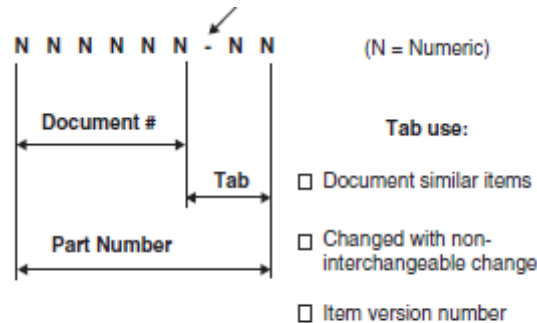
The Product Operational Information consists in the procedures and technical info needed by the operators, and support staff to keep on working, maintain and dispose the product.

The identification is the sum of all features that describe a unique product

Part Number Identification

Golden Rules

- a. Every product has an identifier (Part Number) chosen by its developer.



Rule of thumb for drawing title block:
 00 for doc only, 01 first part, if tabulated XX

- b. A PN used with no change in a new development maintain its identifier.
- c. If the product changes, the identifier changes to indicate that new configuration is not interchangeable with the old configuration (FIT), is physically different (FORM) or has different capabilities (FUNCTION).
- d. If a component of an assembly changes, the assembly identifier has to change (monoconf) or the structure has to reflect the change (multiconf)
- e. SW changes demand careful re-identifications. It can be limited to SW or even affect the HW.

The Part Number of an element

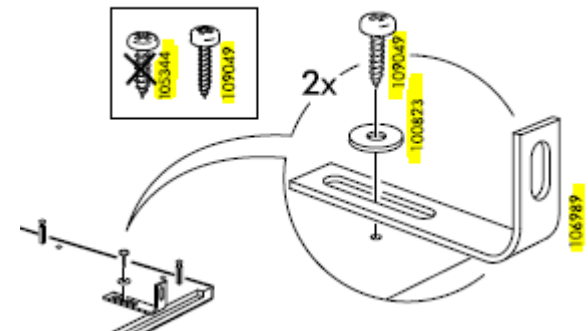
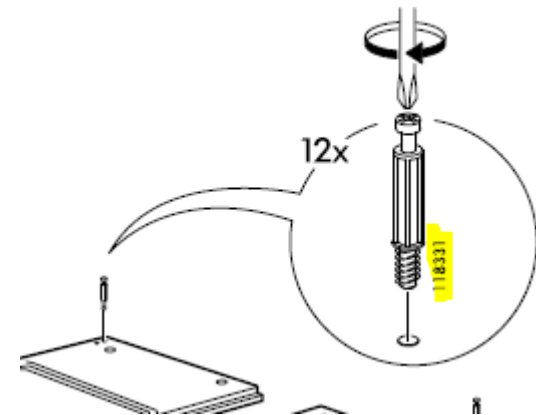
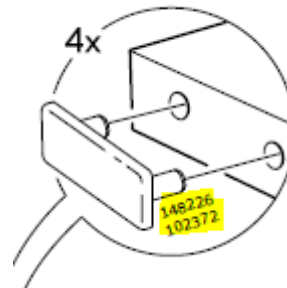
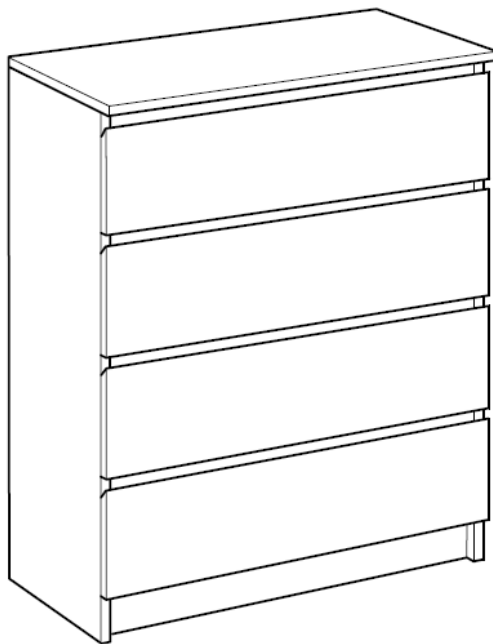
SIGNIFICANT	VS	NON SIGNIFICANT
Describes Part	•	No Significance
Is a Class Code	•	May Need Separate Class Code
Must Publish Code Book	•	No Code Book (or Code Book in CM)
Misinterpretation Likely	•	Eliminates Interpretation
Security May Be Breached	•	Security Better
Often Variable Length and Alpha Numeric	•	Uniform Length Numeric
Less Compatible with MRP / Info Systems	•	More Compatible with MRP, etc.
Check Digit Use Not Practical	•	Lends Itself to Check Digit
Longer Harder to Memorize	•	Shorter Easier to Memorize
More Error Prone	•	Fewer Errors
Probably Separate Document Number	•	Document Number Part of the Part Number
Categories Usually Breakdown - Limited Life	•	No Categories Lasts Longer

The Lego Example: PN Identification

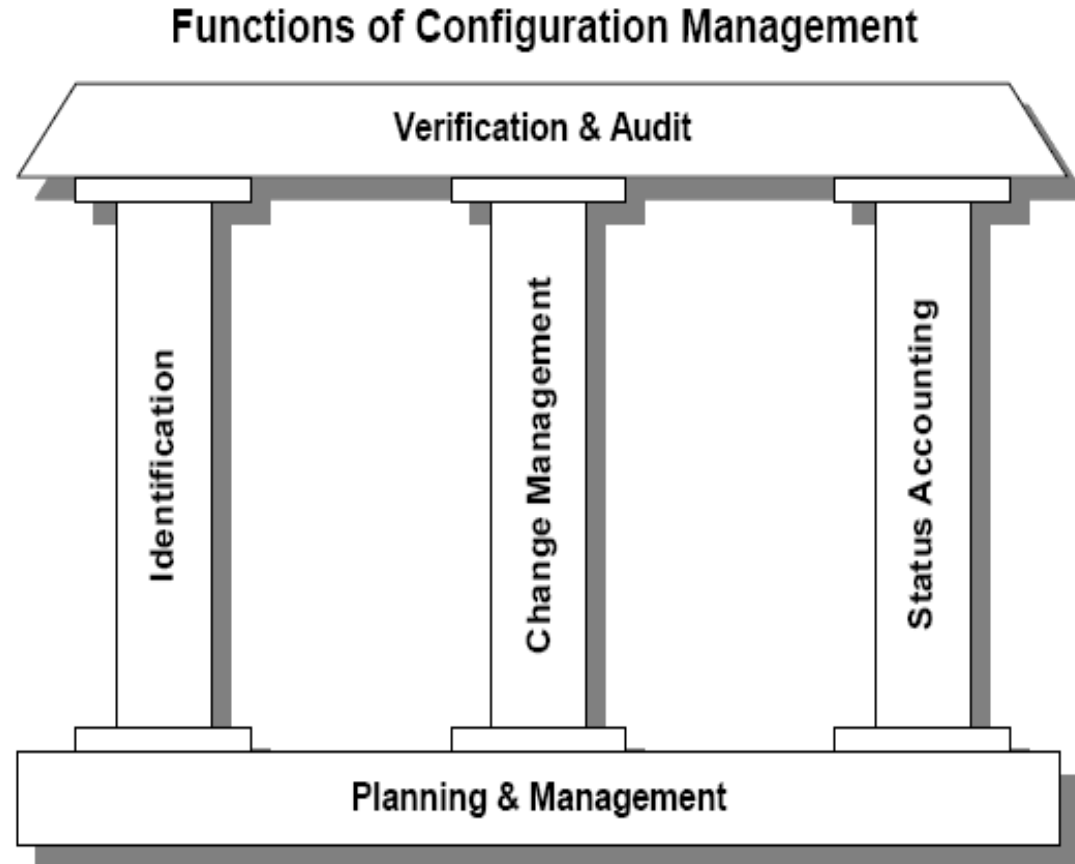


Example of a PN
identification

MALM



1. CM Planning and Management
2. Configuration Identification
3. **Configuration Change
Management**
4. Configuration Status Accounting
5. Configuration Verification and
Audit



A baseline is an agreed description of the characteristics, attributes, components and defining elements that constitute the defined item.

The configuration baselines are used to define and describe the characteristics of a product in a given moment.

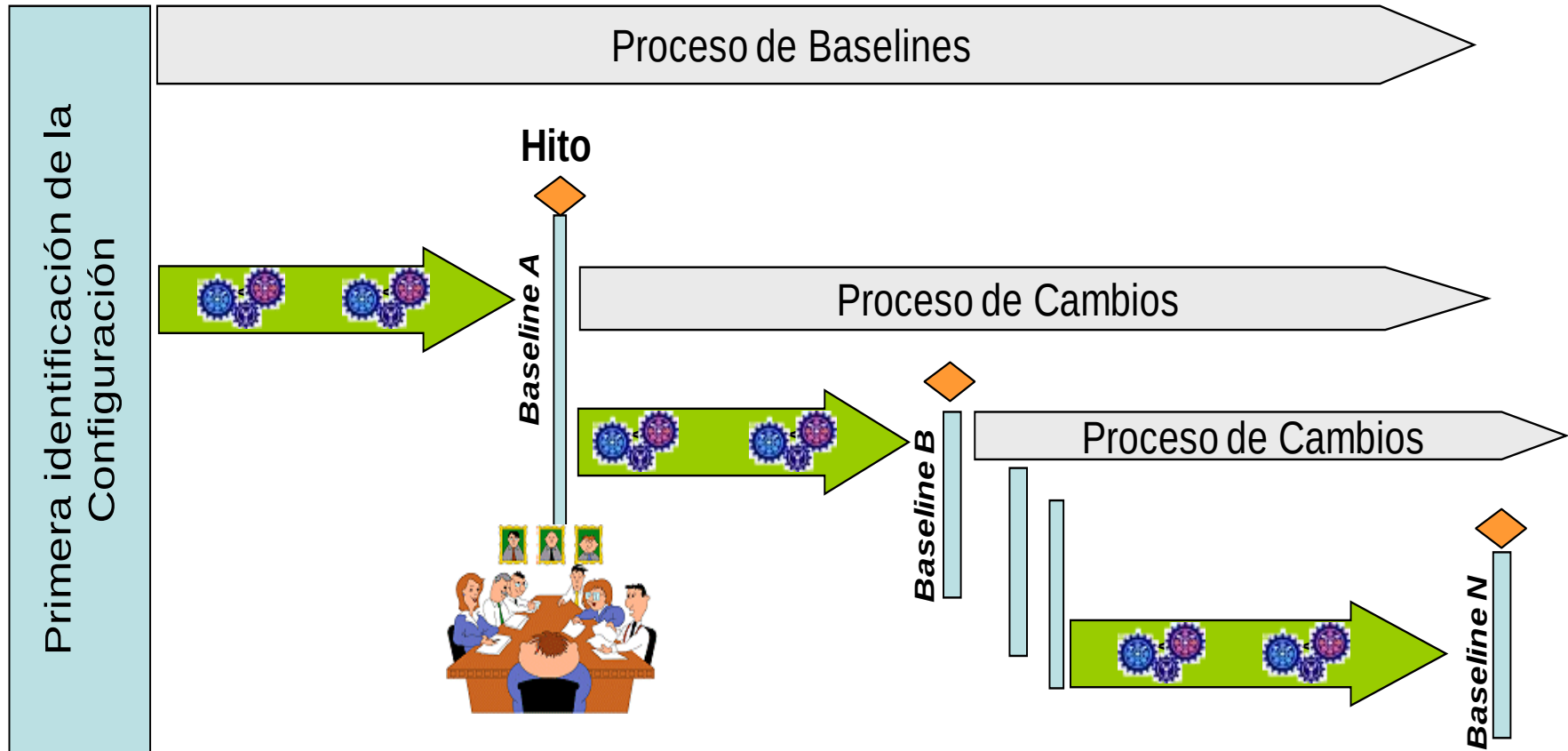
- A Baseline can be used to certify the performance/characteristics of a product
- A Baseline can be used as a reference point for change management in a later stage. Sometimes, change management is reduced to baseline management.

A baseline can be simple and used for internal use or formal, detailed and developed under strict rules.

The baselines are recorded and maintained in order to make possible the retrieval of the configuration in a given moment and the reasons that lead to its evolution.

A baseline are key for sharing interfaces between stakeholder developing a product within a project.

Baseline Evolution



A change is any alteration of the content of an approved baseline or a given definition.

The Change Management function includes the control of the changes and the variations of a product using a systematic and measurable process independent of the controlled product or its lifecycle.

The objectives of Change Management are:

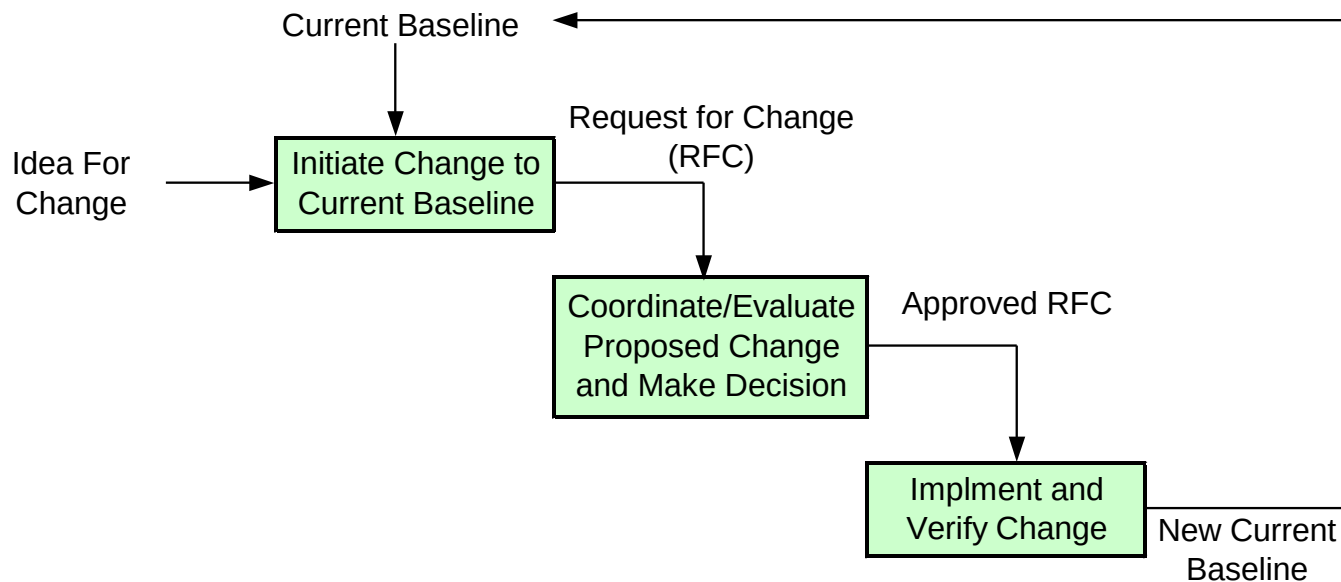
- Identify the need of a change
- Define the change
- Document the change
- Handle and communicate the repercussions of the change
- Evaluate and coordinate the change
- Verified the incorporation of the change and its consistency
- Control, archive and be able to retrieve the overall process and the resulting information

The Lego Example: Change Management



Modifications and Change Process

The Configuration Change Management includes the control of changes and variations of a product by using a systematic and measure process, versatile for different types of products and the phase in its lifecycle.

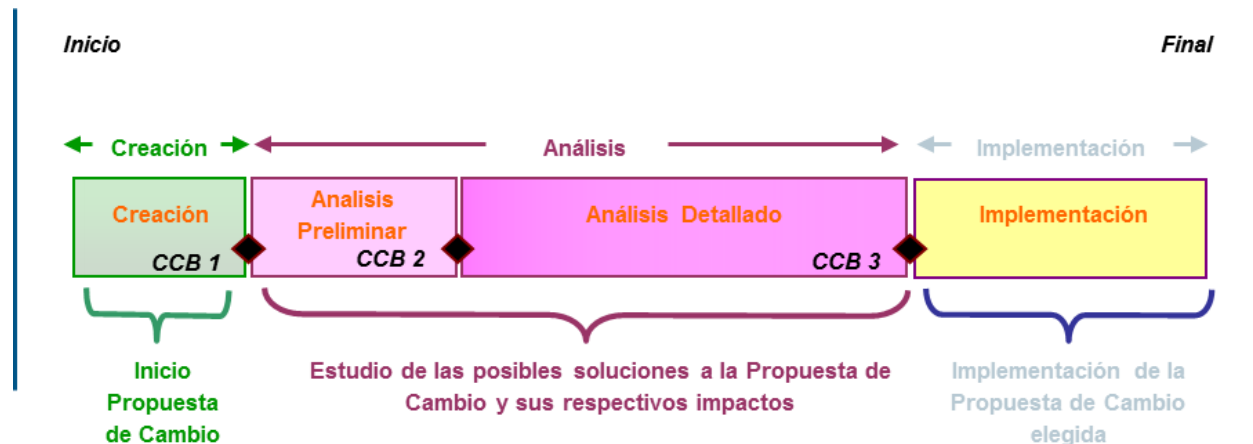


The codes driving the change in a product and making it retrievable have different names: 'Request for Change', 'Modification Proposal', 'Design Change', 'Change Request', 'Mod', 'Authority' etc

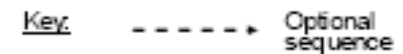
Change Control during Product Life Cycle

Generally includes different sub processes :

- Change **Initiation**: The proposal shall evaluate benefits, advantages, risks, etc and determine different solutions.
- **Evaluation** and impact coordination: repercussions are evaluated and a GO/NO GO (and forecast implementation) decision is taken.
- **Implementation and verification**: Proper actions are taken to assure that the change is implemented in product and documentation



Airframe Tech
DESIGN
L 2 p. 27



Airbus Amber

A Configuration Management process has to be robust enough to involve the different functions in the company affected by a change.

Nevertheless which function initiates the change, the process needs to be able to identify potential impacted to provide the repercussions.

The final approval of a change usually is performed in a committee (the Configuration Control Board), convoked in a regular basis, chaired by the Program Organization and with all major potential actors gathered to ratify the decisions taken by the committee.

The business case of the change is taken into account for its decision and , as well as the GO/NO GO for the change, the decision taken involves the effectivity of the change.

Some other committees can exist along the company to pre-assess the changes proposed in order to early reject those not proper.

The Lego Example: Configuration Control Board



Components in a Configuration Control Board

CCB 1 – CCB 2

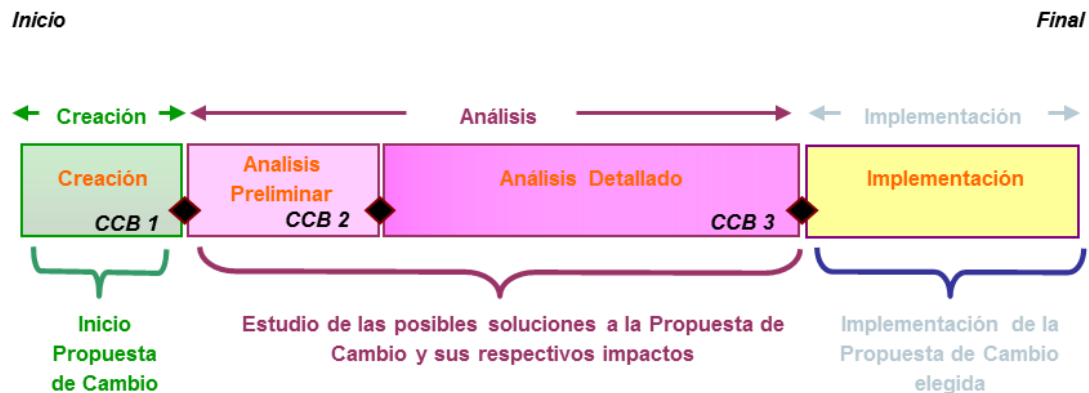
Permanent Actor	Role
Programmes	Programs Configuration (acting as President and secretary).
Engineering	Chief engineering, configuration and design integrator.
Operations	Configuration “as planned” (represented by Manufacturing Engineering).
Services	Configuration in Service and Service Programs/Integrator.
Procurement	Procurement Focal Point

Actor when required	Role
Quality	Quality Focal Point
Financial	Responsible for financial area represented by Program Controllers.
Other	Any other organization involved or interested in the change proposal.

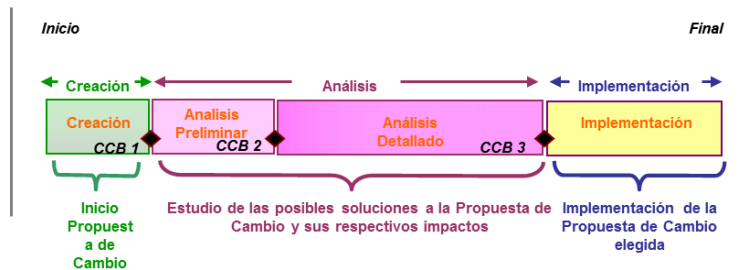
CCB 3

Permanent Actor	Role
Programmes	Configuration of Programs will lead the CGM and will act as secretary.
Engineering	Engineering Directorate and configuration.
Services	Responsible for Configuration Management in Services Responsible for Programmes in Services.
Operations	Configuration “as planned” in Operations (Represented by Manufacturing Engineering).
Procurement	Focal Point of Procurement.
Quality	Responsible for Configuration Management area or Quality Changes Coordinator.

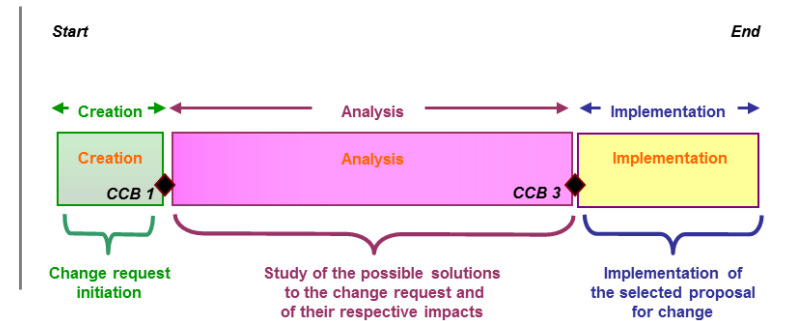
Actor when required	Role
Financial	Responsible for the financial area represented by Program Controllers.
Engineering	Integrators.
Other	Any additional function of the permanent actors or other organization involved or interested in the change proposal.



A change process has to be capable to deal with last minute changes, simplified documentation and fastest approval cycles. All these methods have to be validated and included in the process and usable for certain situations.



STANDARD PROCESS



FAST TRACK

The fast track solutions can affect:

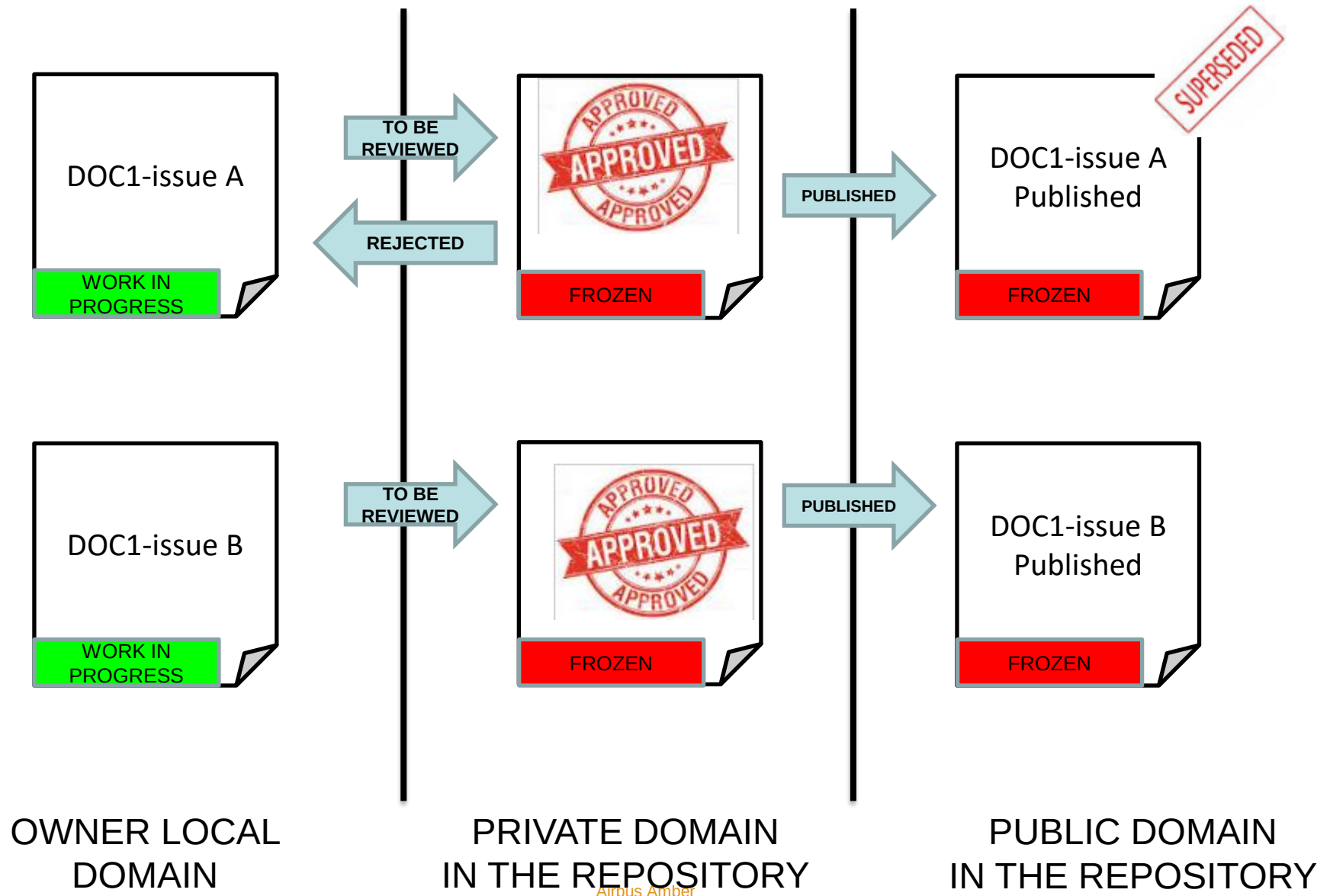
- The change process: less decision gates, less maturity states.
- The documentation: validated simplified documentation.
- The Release and approval process: simplified signature loops, signature delegations.
- The Change approval itself: Special modification categories, faster channels of configuration control board approval (approval between committees).

The proper control of the documentation demands the following aspects:

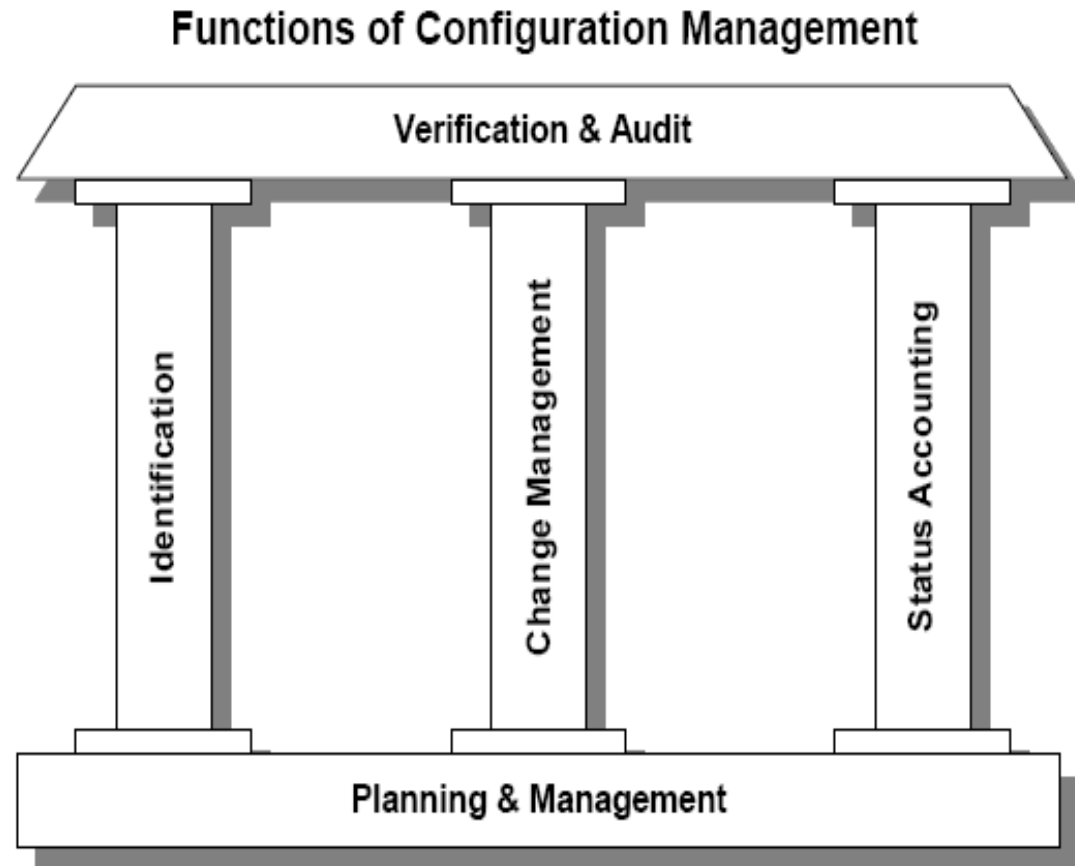
- **Status in elements.** Each element needs an assigned value due its maturity ('in work', 'in review', 'published', 'cancelled')
- **Visibility.** The whole authorized community have to be able to know the status of the elements
- **Capability of reissue.** An element has to be able to be re-issued
- **Retrieval.** The superseded issues shall be accesible, as well as their signatures data, dates, etc

This shall be reflected in the procedures and introduced in the tools for managing the documentation,

Document release example



1. CM Planning and Management
2. Configuration Identification
3. Configuration Change Management
4. **Configuration Status Accounting**
5. Configuration Verification and Audit



The configuration status accounting (CSA) is a function that secures the capability of retrieving easily and safely the information that defines a product along its lifecycle

The purpose and main advantage of a CSA is to capture and keep the configuration info in order to assure that current and legacy configurations can be successfully retrieved.

A CSA shall :

- Keep record on the info along all the lifecycle
- Be able to retrieve the info to permit investigations on Design troubles, design changes, decisions on changes, audits, etc
- Authorize the access to the info taking into account the user
- Track the historical configuration of the product
- Be able to foster process improvements

The industry is plenty of different products covering the product lifecycle development. Some of the COTS tools in the market are:

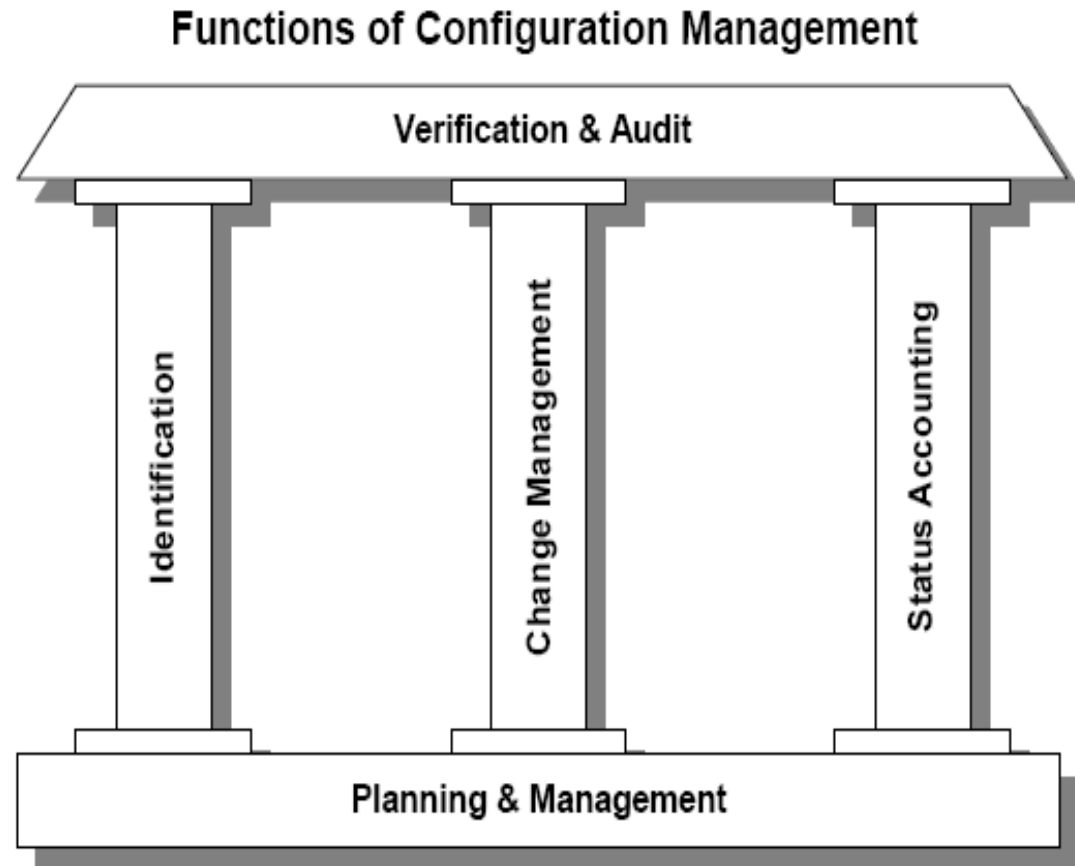
- PTC Navigate (PTC)
- Windchill (PTC)
- Team Center (Siemens)
- CATIA V6/3D Experience (Dassault)
- Agile PLM (Oracle)
- SAP PLM

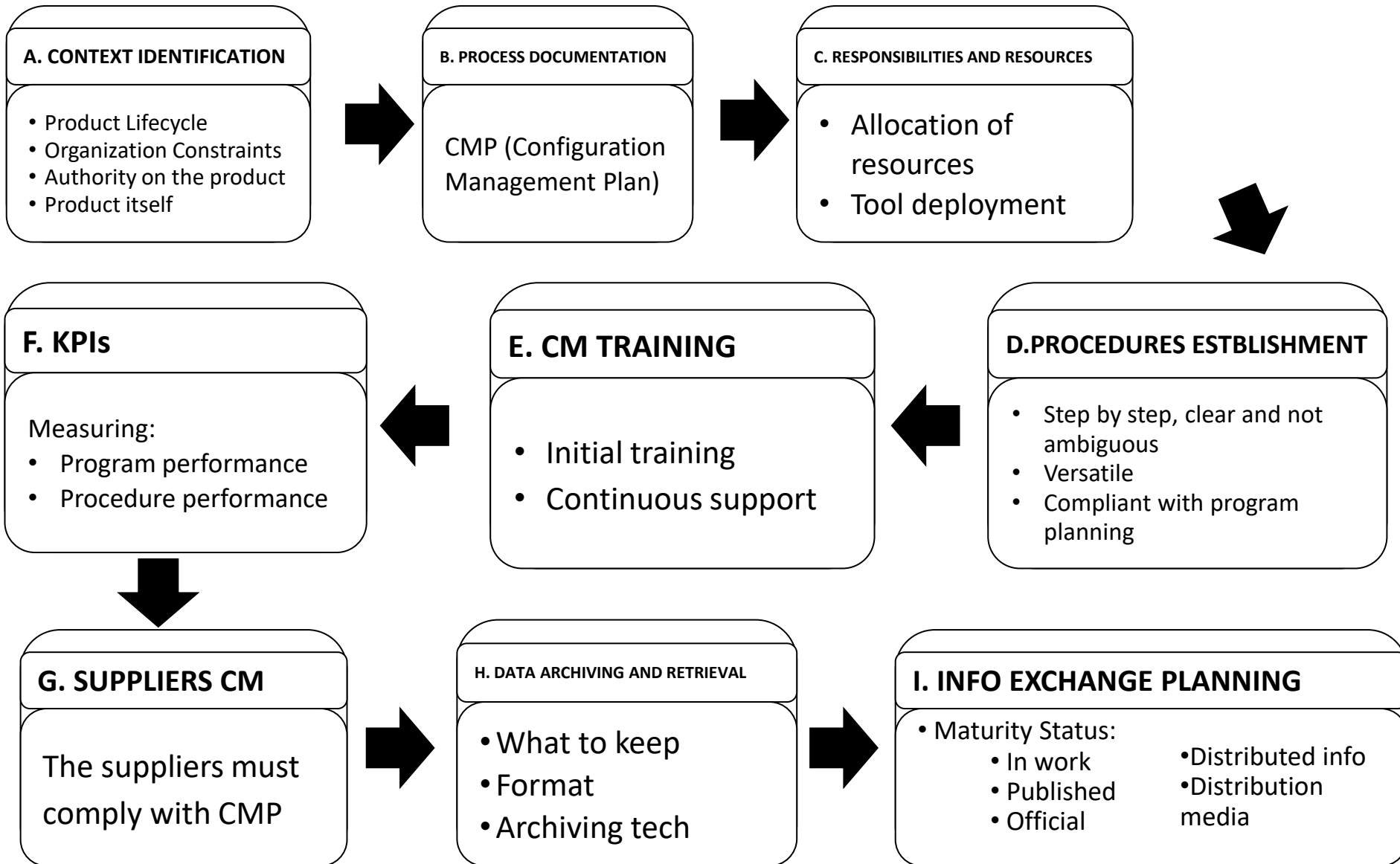
Also, many other companies decide to incorporate their configuration requirements into in house developed tools. The advantage lies in a higher freedom to tailor the processes.

The current trends tend to:

- Full lifecycle tools covering the whole process in less tools
- Mixed solutions: COTS + In house for different functions

1. **CM Planning and Management**
2. Configuration Identification
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Management
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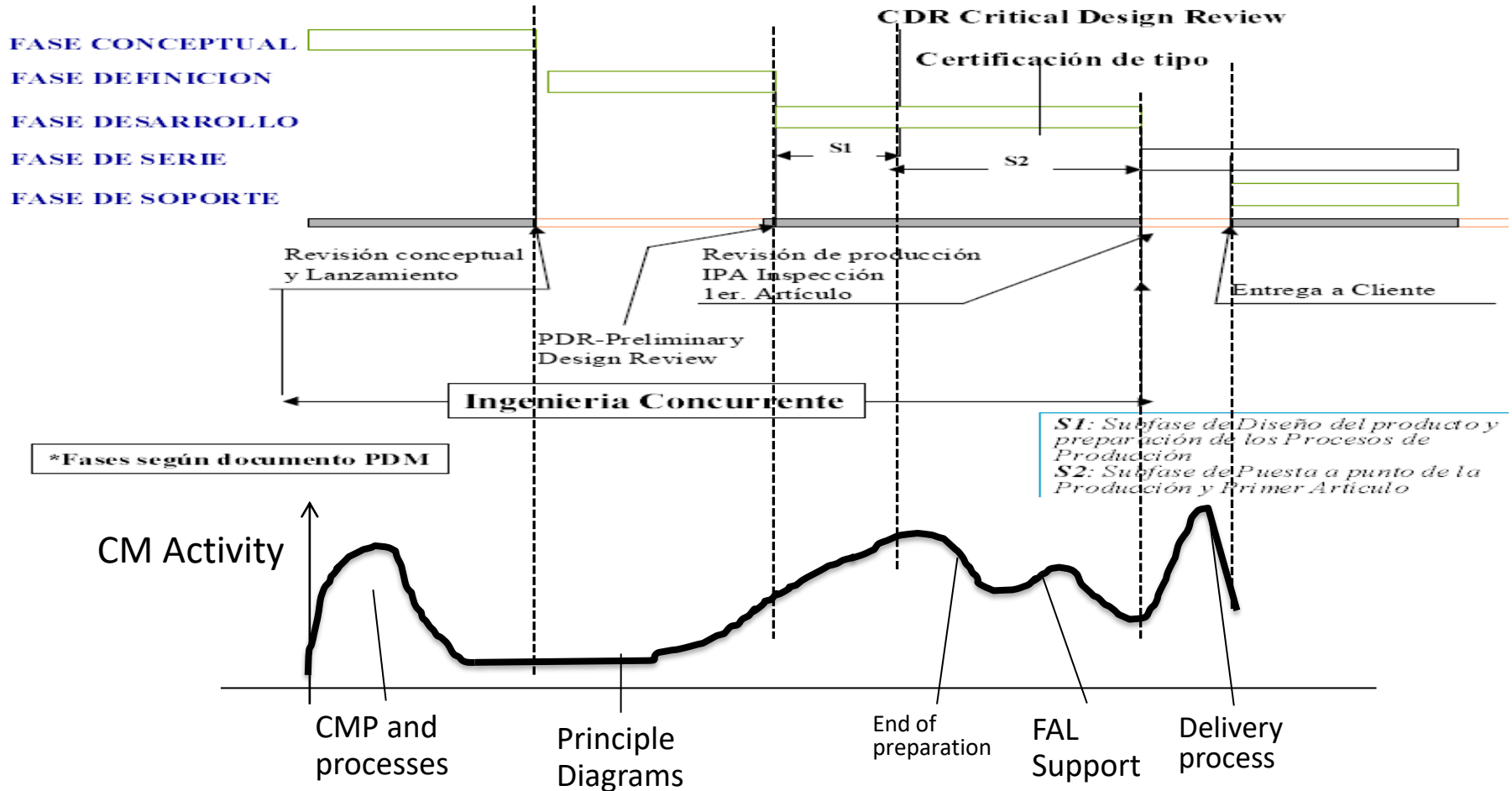


Project Lifecycle



Configuration Management during different phases

Phases >	Conception	Definition	Build	Distribution	Operation	Disposal
Aliases	Concept Exploration Marketing Predevelopment Research Study Material Solution Analysis Technology Development	Design Development Engineering Engineering & Manufacturing Development Program Definition Source Coding Software Build	Construction Fabrication Manufacturing Production Software Build	Delivery Deployment Fielding Installation Sales	Maintenance On orbit Operation & Support Operational Performance Repair Service Life Warranties In-Service	Decommissioning Disposition Removal from Service
Characteristics Of the Phase	Analysis Functions Investigation Need Opportunity Mission Pre-Concept & Concept Definitions Survey Trade-Offs Requirements Definition	Architecture Detailed Design Evaluation Manufacturing Planning Preliminary Design Prototyping Software Code & Test Specification System Definition Testing Functional and Allocated Baselines	Assembly Facility Construction Installation Inspection Manufacture Production Product Baseline	Acceptance Commissioning Deployment Installation Order Setup Stock Supply Transport	Depreciate Maintain Modernize Modify Operate Recover Sustainment Upgrade Use Utilization	Archive Deactivate Destroy Disassemble Discard Disposition Environmental Impact Historical Significance Legacy Mothball Recycle Retrieval Scrap



Configuration Management during different phases

	Concept	Definition	Development	Series	Services Support
Activities	- CMP preparation - Rules, standards and procedures applicable - Identification Rules	- CMP issue - Dedicated standards - Process, materials and standards selection	- Product structure and related documentation - Equipment structure or equipment list in advance	Product structure updates due to series changes	Changes introduction through Service Bulletins
Change Management	NO	It can be controlled	YES	As in development	SB can be based in mod info and authorized through Design Changes
CM Data registry	NO	NO, but preliminary data according standard	YES	As in development	Services Tools
Baseline	Kick Off Documentation	Doc for PDR	Doc for CDR	Certification and Permit to Fly baselines	Services Tools
Verification and audit	NO	NO	SI	SI	NO

There is an existing specialty and more widely spread in the industry dealing with Software Configuration Management.

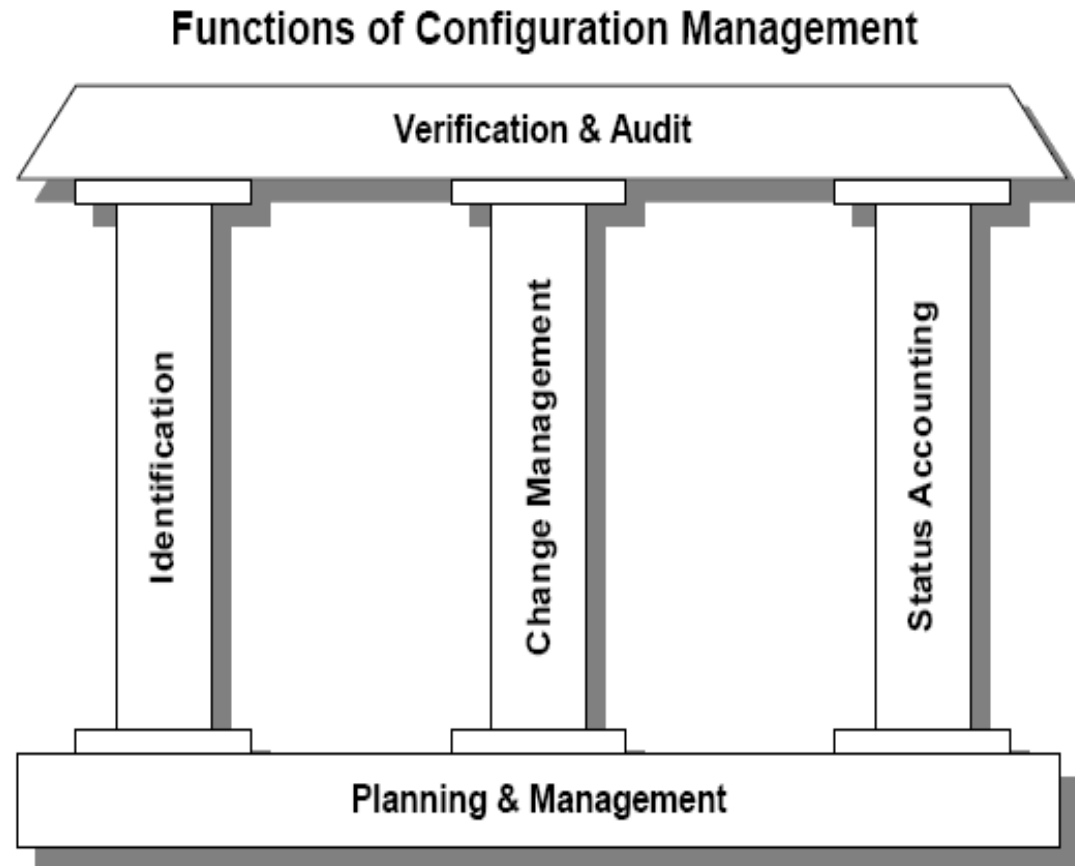
The same concepts (changes, baselines, structures, release process) can be applied to Software Configuration Management.

The main differences are:

INDUSTRIAL HARDWARE	SOFTWARE
Need of dedicated tooling and industrial facilities	Smaller capital to develop a project
Difficult of changing requirements once approved	Possibility of agile change of requirements
High cost in produced item	Capable of offering actual alternatives on the same demos
Slower development	Faster development

Industrial projects are looking in software development to capture the advantages of quicker development (for example with new techniques like Additive Layer Methods)

1. CM Planning and Management
2. Configuration Identification
3. Configuration Change Management
4. Configuration Status Accounting
5. **Configuration Verification and Audit**



The verification and audit of the configuration is the function that establishes that the operation and the functional requirements defined initially have been met by the Design and that the Design has been documented exactly in the info for product definition.

The audits are used to test and validate the processes. Even the configuration management process has to be subjected to an audit, review and enhancement cycle.

There are two different typical examples of configuration audits

- The **Functional Audit**: It is the formal test to verify that a given function has reached the performance characteristics specified in the Definition and Configuration Documentation.
- The **Physical Audit**: It is the formal test to verify that a physical characteristic has reached the performance characteristics specified in the Definition and Configuration Documentation.

PHYSICAL TESTS IN PRODUCTS

A design of a recently developed product or a change made to that product have to be verified to check if the specs, requirements and functional performance are achieved.

The verification methods are planned carefully, efficiently and with utility criteria.
Ex: Planning for destructive tests in test specimen.

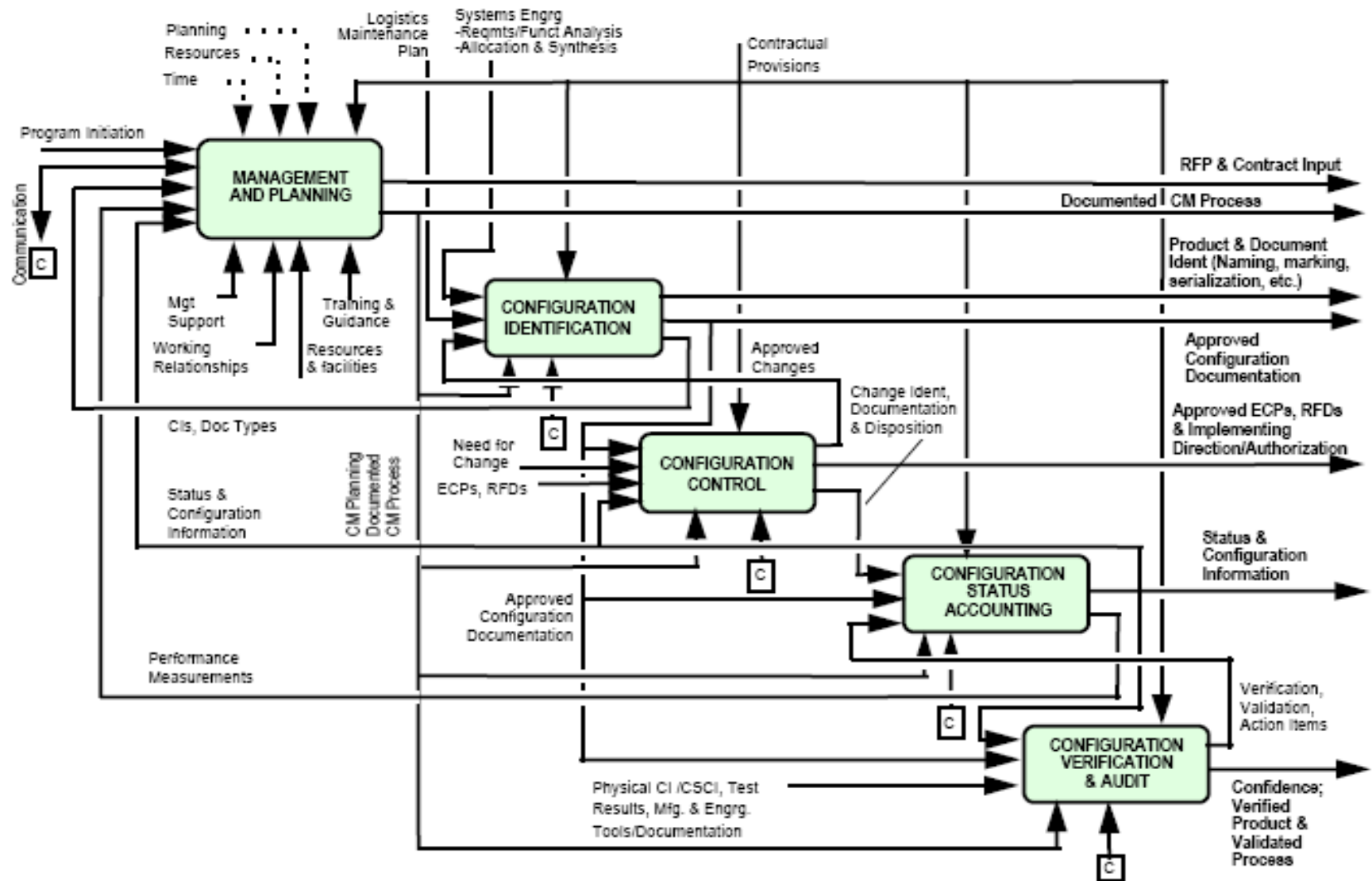
INFO AUDITS IN DOCUMENTATION

The characteristics of a product have to be properly (with exactitude) in the Design Documentation.

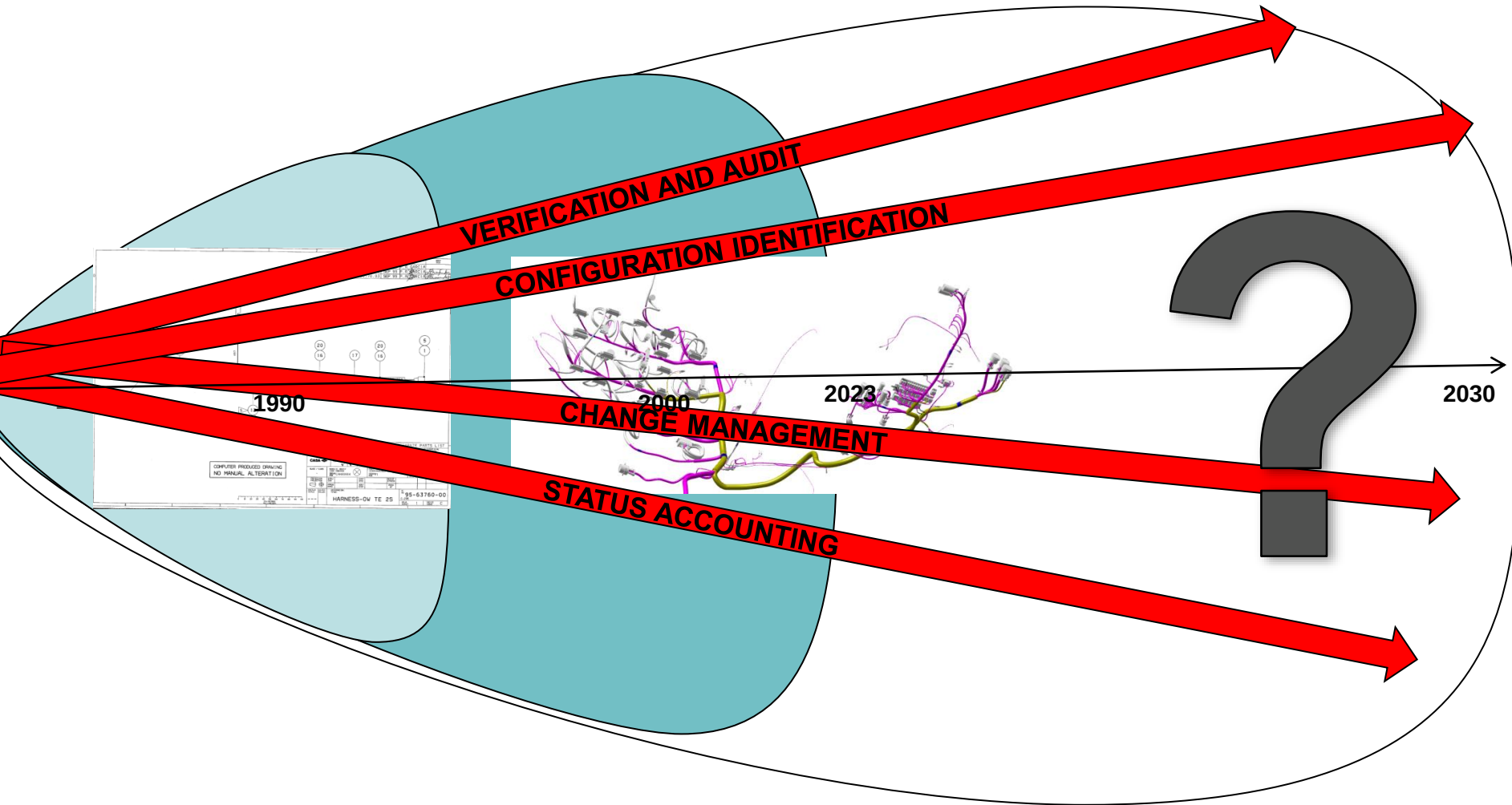
It can be also a matter of audits the traceability of the Design Documentation and the Operational Documentation

Both process are driven independently and determine the acceptance of a product

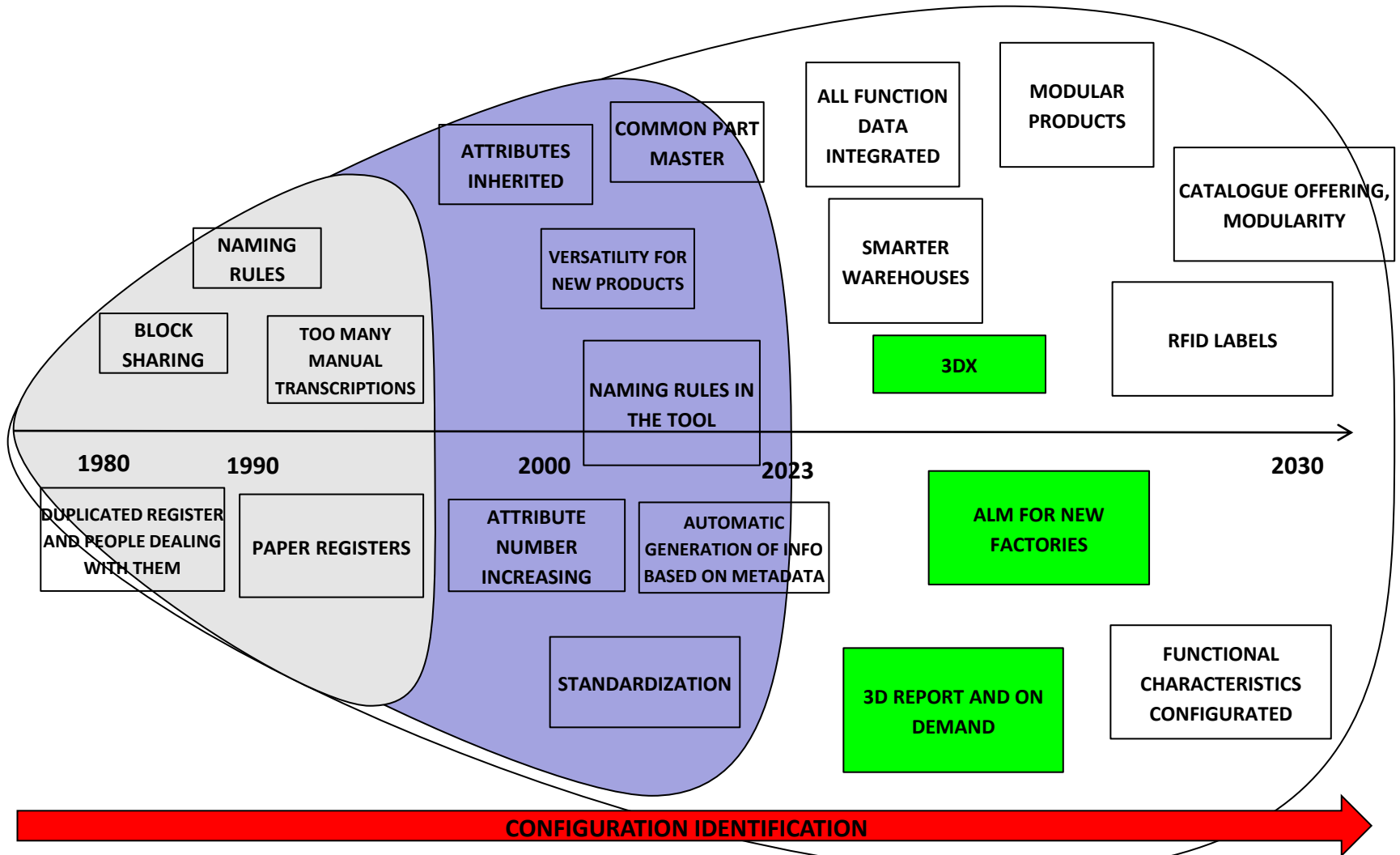
The System Model

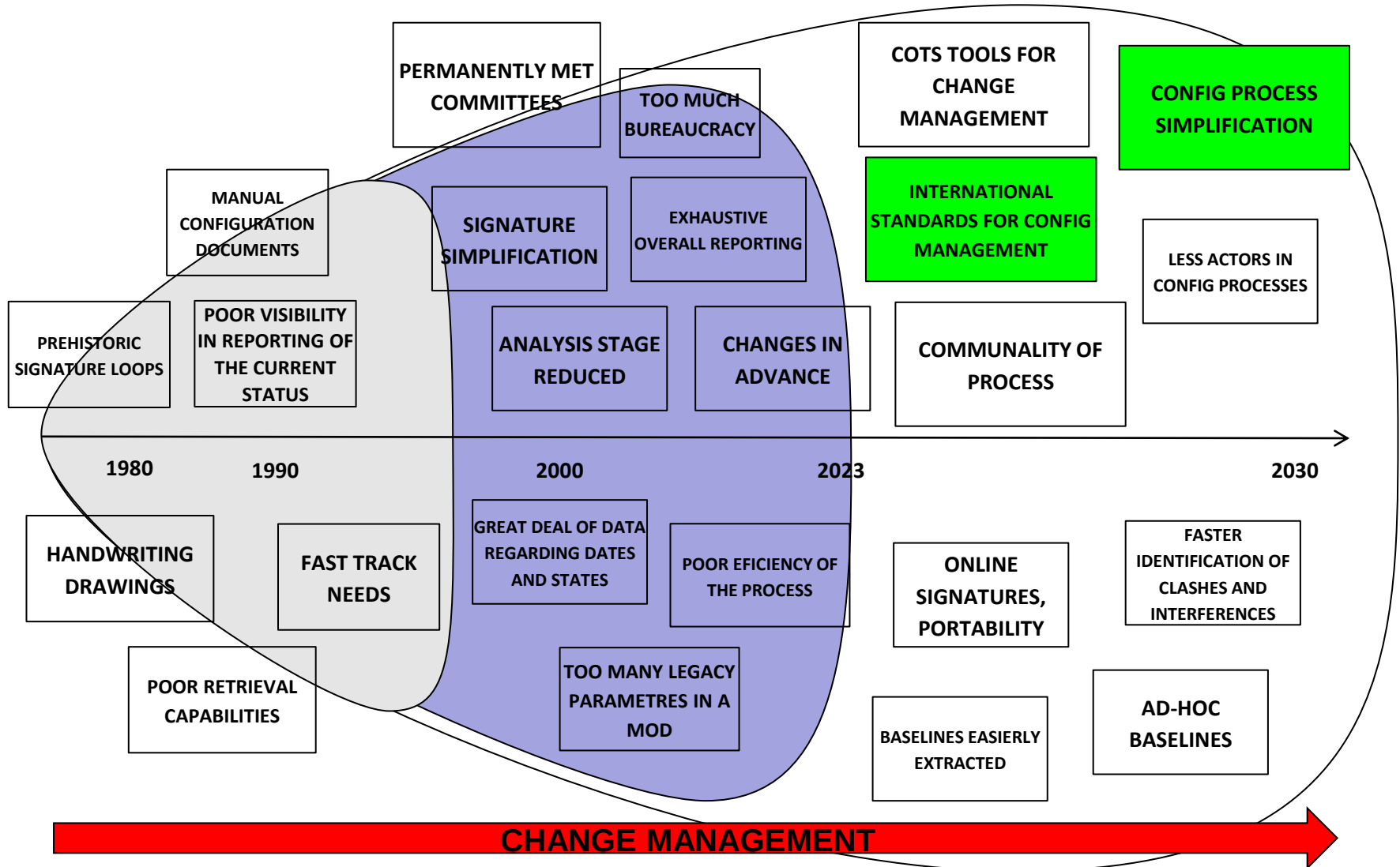


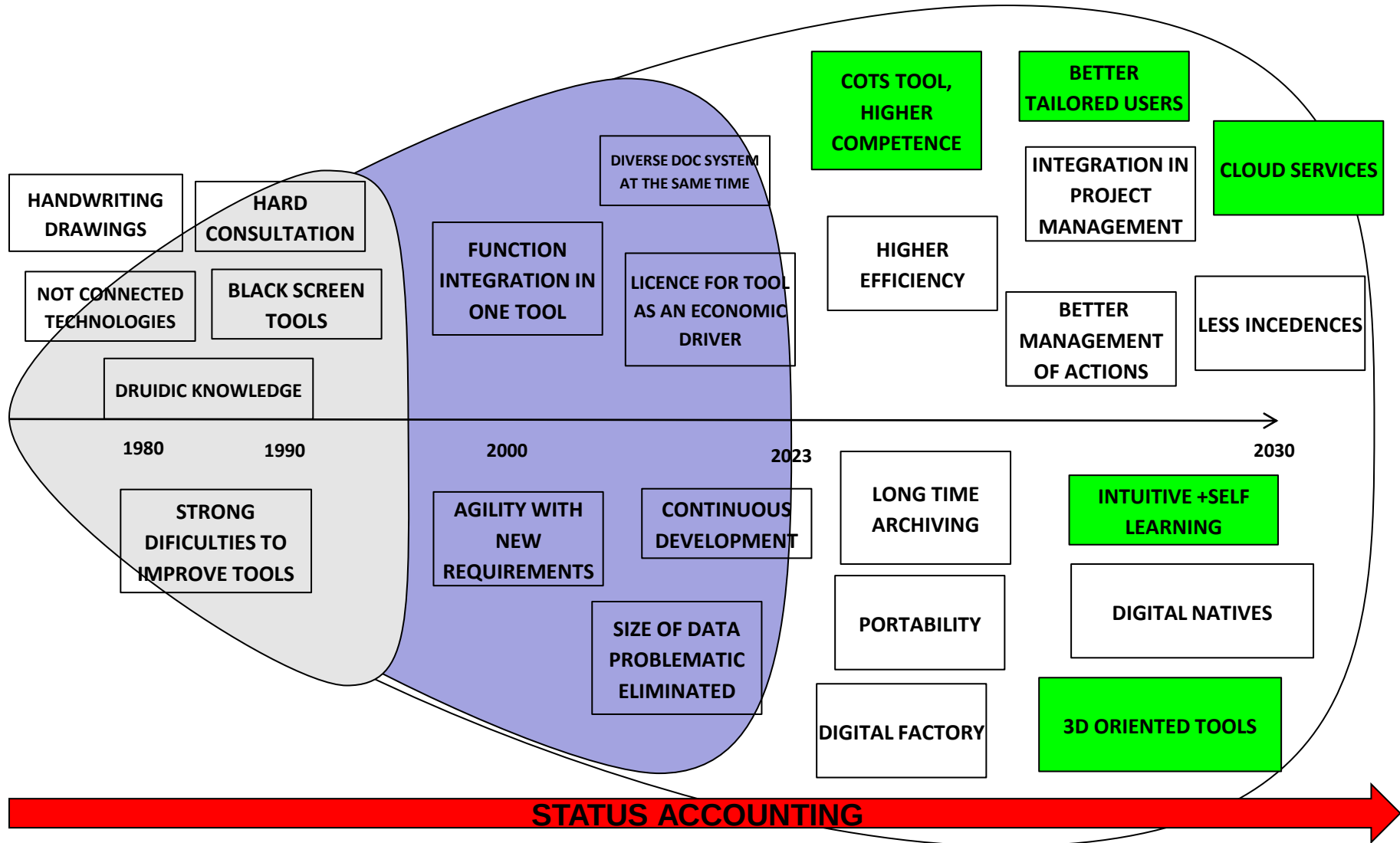
Future trends on Configuration Management



Future Trends on Configuration Identification







Questions and Answers



Configuration Management Standards

1. EN9200. Program Management- Guidelines for Project management
2. ISO10007. Quality Management Systems – Guidelines for Configuration Management
3. EIA-649-C. Configuration Management Standard (Military Products)

Books

1. Watts, Frank B., 'Engineering Documentation Control Handbook. Configuration Management in Industry'
2. Gitlow, Howard S., 'A guide to Lean Six Sigma Management Skills'